Module 2- Exercise

I trust that you now have a solid understanding of the fundamental concepts of medical image analysis, particularly image classification and semantic segmentation. Building on this foundation, you are now expected to implement the following task and submit an individual report for evaluation.

Since these experiments require GPU resources, I encourage you to work in groups. However, each student must implement a different deep learning model from those used by the other members of the group. While you may collaborate on setting up the dataset, preprocessing, and experimental environment, the implementation, analysis, and report must be your own.

Each student is required to submit an individual report. Your report should differ from those of your group members in terms of:

- the model you implemented,
- your experimental analysis,
- your interpretation of the results, and
- your critical reflection on the strengths and limitations of your approach.

Task 1: Polyp Characterization

Perform polyp characterization using one of the following deep learning models (or another model approved by the instructor):

- ResNet (ResNet18, ResNet50, ResNet101, or ResNet152)
- DenseNet
- EfficientNet
- MobileNet
- Vision Transformers (ViTs) (if computational resources permit)

1. Model Description: Begin your report with a description of the model you selected. For example, if you choose MobileNet, your discussion should include:
   - the motivation behind MobileNet,
   - the overall network architecture,
   - the key architectural components (e.g., depthwise separable convolutions),
   - how the model works, and its advantages and limitations.
2. Model Implementation: Implement your selected model for the polyp characterization task. During training, make sure to implement Early Stopping with a patience value of 5 (or another reasonable value if justified).

   You may choose to:
   - train the model from scratch, or
   - fine-tune a pretrained model.

   Clearly state which approach you used.

Table 1: Results.

Model	Pretrained 'or' Scratch	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 Score	AUC
model (Yours)	Scratch					
model (member2)	Pretrained					
model (member 3)	Pretrained					

3. Performance Evaluation: Report the performance of your model and compare it with the models implemented by your teammates. Create a table similar to the Table 1.

4. Additional Results: Include the following evaluation figures for all three models implemented by your group:

   - ROC curves
   - Confusion matrices

5. Discussion and Reflection: Your report should primarily focus on the model that you implemented. In addition, it should include:

   - A comparison between your model and the models implemented by your teammates.
   - An explanation of why certain models performed better or worse.
   - A discussion of the strengths and limitations of your chosen model.
   - A brief reflection on the challenges you encountered, what you learned during the implementation, and possible directions for future improvement.

Task 2: Semantic Segmentation

If you have access to sufficient computational resources, you can use the polyp segmentation dataset available on Nextcloud. Otherwise, you can work with the following open-source Membrane dataset.

Perform semantic segmentation using one of the following deep learning models:

- U-Net
- U-Net++
- PSPNet
- Attention U-Net
- TransUNet

1. Model Description: Provide a description of the model you selected. For example, if you choose U-Net++, your discussion should include:

   - the motivation behind the model,
   - the overall network architecture,
   - its key architectural components,
   - how it differs from the original U-Net, and
   - its advantages and limitations.

2. Loss Function: Select one of the following loss functions for training your model:

   - Dice Loss
   - Dice + BCE Loss
   - Binary Cross-Entropy (BCE) Loss
   - IoU + BCE Loss
   - Mean Squared Error (MSE) Loss

   Clearly justify why you selected the chosen loss function.

3. Model Training: Train your model using the selected loss function. Implement Early Stopping with a patience value of 5 (or another reasonable value if justified).

4. Performance Evaluation: Compare your model with the segmentation models implemented by your group members. Report the following performance metrics as in Table 2. Since the Membrane dataset does not provide ground-truth masks for the test images, please report the evaluation metrics only on the training set. If you are using the AICE dataset from Nextcloud, you can report the metrics on the test set as well.

Table 2: Results.

Model	Jaccard	F1 Score	Recall	Precision	Accuracy
Dice Loss					
your model					
member1 model					
member2 model					

5. Qualitative Results: Include the segmentation results for at least two test images. For each example, present:

   - the original image,
   - the predicted segmentation.

6. Discussion and Reflection: Your report should primarily focus on the model that you implemented. In addition, include:

   - A comparison between your model and those implemented by your teammates.
   - A discussion of why some models performed better than others.
   - An analysis of the effect of the selected loss function on segmentation performance.
   - A critical reflection on the strengths and limitations of your approach.
   - A brief discussion of what you learned during the implementation and possible future improvements.